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The Editors-in-Chief
International Review of Financial Analysis (Elsevier)

Re: Submission of Original Research Article "Causal Emergence in Financial Markets: Dynamic Organization and Effective Dimensionality During Systemic Stress"

Dear Editors-in-Chief,

I am pleased to submit my original research manuscript titled **"Causal Emergence in Financial Markets: Dynamic Organization and Effective Dimensionality During Systemic Stress"** for publication consideration in the *International Review of Financial Analysis* (IRFA).

A central question in empirical financial economics and systemic risk analysis is whether collective financial market distress develops an organized macroscopic dynamical structure that cannot be reduced to contemporaneous covariance, common factor exposure, or univariate persistence.

In this study, I operationalize continuous-state *Causal Emergence* (CE) across daily U.S. equity industry portfolio returns from January 1990 to June 2026. By modeling cross-sector transitions via Vector Autoregressions and optimizing coarse-graining projection matrices on Stiefel manifolds under scale-adaptive Gaussian interventions, I introduce: (1) the **Causal Emergence Financial Index** (CEFI_t), quantifying macroscopic effective information density gained by coarse-graining; and (2) the **Causal Effective Dimension** (q_t^*), identifying the macro dimension concentrating dynamic structure.

To ensure econometric defensibility and prevent mechanical optimization bias, I evaluate historical estimates against a four-tier hierarchy of matched surrogate null models ($B = 9,999$ for primary nulls). Key empirical findings include:

- **State-Dependent Emergence:** Across historical stress episodes, observed emergence significantly exceeds static correlation and cross-lag isolation nulls ($p_{\mathrm{Holm}} \leq 0.0396$).
- **Cross-Lag Organization:** During both the 2008 GFC peak and the 2020 COVID crash trough, observed emergence significantly exceeds the cross-lag network isolation null ($p_{\mathrm{emp}} = 0.0001, p_{\mathrm{Holm}} = 0.0006$), reflecting structured intertemporal coupling across sectors.
- **Dimensionality Concentration:** Systemic stress dynamically concentrates causal transitions into low macroscopic dimensions ($q^* \leq 2$ in over 93% of historical windows, modal $q^* = 2$).

The manuscript has not been published previously, is not under consideration elsewhere, and its submission is approved by the author. I confirm compliance with all ethical and editorial policies of the journal.

Sincerely,

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